

Below are structured theoretical notes with detailed explanations and authoritative Microsoft Learn references for each requested topic. You can compile them into study material for Azure AI – Fundamentals.

1. Create & Manage Microsoft Azure Resources

Theory & Explanation:

Manage Azure resources via Portal, CLI, PowerShell, SDKs, ARM/Bicep, Terraform, and Ansible. Use Azure Resource Manager (ARM) to deploy, update, delete, and apply RBAC, tags, and locks. Guidance includes grouping by Resource Groups, subscriptions, and Management Groups, using templates for consistent deployments. [\[learn.microsoft.com\]](#), [\[learn.microsoft.com\]](#), [\[learn.microsoft.com\]](#), [\[learn.microsoft.com\]](#)

2. Create an Azure AI Resource

Theory & Explanation:

Provision Azure AI resources (OpenAI, Foundry) via Azure Portal, CLI, PowerShell, ARM/Bicep. You set subscription, resource group, region, name, pricing tier, networking, and tags. Foundry offers organization of AI projects across hubs, projects, and deployments. [\[learn.microsoft.com\]](#), [\[learn.microsoft.com\]](#), [\[learn.microsoft.com\]](#)

3. Configure Diagnostic Logging

Theory & Explanation:

Use Azure Monitor's diagnostic settings to collect resource logs, activity logs, and platform metrics. Destinations include Log Analytics, Storage Accounts, and Event Hubs. Can manage at scale via Azure Policy initiatives. For Foundry Tools, enable `Audit`, `RequestResponse`, `AllMetrics` with storage and Log Analytics. [\[learn.microsoft.com\]](#), [\[learn.microsoft.com\]](#), [\[learn.microsoft.com\]](#), [\[learn.microsoft.com\]](#), [\[learn.microsoft.com\]](#)

4. Manage Costs for Azure AI Services

Theory & Explanation:

Apply FinOps principles: planning, visibility, optimization. Use Azure Cost Management for budgeting, cost analysis, alerts, and integrated Azure Advisor recommendations. For AI workloads, follow cost management best practices (FinOps, reservation, tagging, architectures) and use Cost Optimization workbook. [\[learn.microsoft.com\]](#), [\[learn.microsoft.com\]](#), [\[learn.microsoft.com\]](#), [\[learn.microsoft.com\]](#), [\[learn.microsoft.com\]](#)

5. Monitor an Azure AI Resource

Theory & Explanation:

Use Azure Monitor to capture platform metrics and logs; includes Overview, Activity Log, Metrics, Alerts, Insights. AI-specific: Cognitive Search, Machine Learning, OpenAI have dedicated monitoring documents detailing telemetry, metrics, logs, and advanced patterns like gateway-based usage

auditing. [\[learn.microsoft.com\]](#), [\[learn.microsoft.com\]](#), [\[learn.microsoft.com\]](#), [\[learn.microsoft.com\]](#), [\[learn.microsoft.com\]](#)

6. Deploy Microsoft Azure AI Services

Theory & Explanation:

Deploy models or applications using Foundry standard, serverless, managed compute, AKS, Container Apps, App Service, or AKS with OpenAI. Supports LLM archetypes, retrieval-augmented generation (RAG), REST API, private endpoints. [\[learn.microsoft.com\]](#), [\[learn.microsoft.com\]](#), [\[learn.microsoft.com\]](#), [\[learn.microsoft.com\]](#), [\[learn.microsoft.com\]](#)

7. Determine a Default Endpoint for a Service

Theory & Explanation:

By default, services expose a public REST endpoint (via region-specific address). To secure, configure Virtual Network Service Endpoints or Private Endpoints. Service Endpoints route public traffic over Azure backbone—use Azure CLI: `az network service-endpoint list`. [\[learn.microsoft.com\]](#), [\[learn.microsoft.com\]](#), [\[learn.microsoft.com\]](#)

8. Integrate AI Services into CI/CD Pipelines

Theory & Explanation:

MLOps and GenAIOps integrate Azure AI services into DevOps pipelines via Azure Pipelines, GitHub Actions, Terraform. Automate model training, deployment, monitoring with Azure Machine Learning SDK v2, pipelines, prompt flow. Use recommended architecture accelerators for stable CI/CD. [\[learn.microsoft.com\]](#), [\[learn.microsoft.com\]](#), [\[learn.microsoft.com\]](#), [\[learn.microsoft.com\]](#), [\[learn.microsoft.com\]](#)

9. Plan Container Deployment / Prebuilt Containers in Connected Environments

Theory & Explanation:

Use Azure AI Foundry Docker containers (Speech, Text, Vision) on-premises or in Azure Container Instances or Apps. Supports disconnected/connected environments. Choose between disconnected (no internet) or connected contexts; configure via Core Container Registry. For orchestration, use Kubernetes/AKS/Azure Container Apps for GPU and scale. [\[learn.microsoft.com\]](#), [\[learn.microsoft.com\]](#), [\[learn.microsoft.com\]](#), [\[learn.microsoft.com\]](#), [\[learn.microsoft.com\]](#)

Each section includes key concepts, configuration steps, and official Microsoft Learn references for deeper reading and validation.